Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using a retail dataset to uncover insights into customer demographics, purchasing patterns, and factors influencing spending behavior. The objective is to support data-driven decision-making by identifying key drivers of revenue, customer engagement, and product performance. The analysis follows an end-to-end workflow, including data preparation in Python, business analysis in SQL, and visualization in Power BI.

2. Dataset Summary

- ***Rows:*** 3,900
- ***Columns:*** 18
- ***Key Features:***
  - Customer demographics: Age, Gender, Location, Subscription Status
  - Purchase details: Item Purchased, Category, Purchase Amount, Season, Size, Color
  - Shopping behavior: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
- Data Quality:
  - Missing Data: 37 values in Review Rating column
  - Handled using median imputation by category

3. Data Preparation & Feature Engineering (Python)

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.

- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

1. **Revenue by Age Group** – Analyzed which age group contributes the most to total revenue and compared their average spending levels.

age_group	total_custom...	avg_spend	total_revenue
Young Adults	1028	60.45	62143
Middle-aged	986	60.04	59197
Adult	942	59.42	55978
Senior	944	59.07	55763

2. **Subscribers vs. Non-Subscribers –** Evaluated whether subscribed customers generate higher revenue and spend more on average compared to non-subscribers.

subscription_sta...	total_custom...	avg_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

3. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount, indicating strong purchasing power despite promotions.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91

4. **Ratings vs. Spending Behavior** – Explored the relationship between product ratings and customer spending to determine whether higher-rated products lead to higher purchase amounts.

item_purchas...	avg_rating	avg_spend
Gloves	3.86	60.55
Sandals	3.84	57.50
Boots	3.82	62.63
Hat	3.8	60.88
T-shirt	3.78	62.91
Handbag	3.78	57.89
Skirt	3.78	59.51
Sweater	3.76	57.70
Sneakers	3.76	59.55
Belt	3.76	59.84
Jewelry	3.76	58.54
Jacket	3.76	56.74
Sunglasses	3.75	59.93
Dress	3.75	62.17

5. **Discount-Driven Products** – Determined which products have the highest percentage of purchases with discounts applied, highlighting items that rely heavily on promotions.

item_purchas...	discount_ra...
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

6. **Customer Segmentation (Loyalty Analysis)** – Segmented customers into New, Returning, and Loyal groups based on their purchase history to understand customer retention and engagement levels.

segment	total_custom...
Loyal	3116
Returning	701
New	83

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.

1. **Key Performance Indicators (KPIs):**

- ***Total Customers*** – total number of unique customers
- ***Total Revenue*** – overall sales generated
- ***Average Purchase*** – average spending per transaction
- ***Customer Engagement*** – measured by the average number of previous purchases

2. **Key Visualizations:**

- ***Revenue by Category –*** identifies top-performing product categories
- ***Revenue by Age Group –*** highlights revenue contribution across demographic segments
- ***Subscription Distribution –*** shows the proportion of subscribed vs. non-subscribed customers
- ***Discount Usage Analysis –*** examines the role of promotions in purchasing behavior
- ***Seasonal Trends –*** analyzes how purchase frequency varies across seasons

3. **Dashboard Insights:** The dashboard highlights several important patterns in customer behavior. Revenue is concentrated in specific product categories, indicating strong product demand. Younger customer segments contribute a significant portion of total revenue. Additionally, a large proportion of customers are non-subscribers, suggesting potential for growth in subscription programs. Discounts also appear to play a key role in influencing purchasing decisions.

6. Key Insights

Based on the analysis, several key insights were identified:

- **Customer Segmentation Matters**

Certain age groups contribute more significantly to total revenue, indicating the importance of demographic targeting.

- **Product Categories Drive Revenue**

A small number of product categories generate the majority of revenue, highlighting core business strengths.

- **Subscription Opportunity Exists**

Most customers are non-subscribers, suggesting untapped potential for improving customer retention and lifetime value.

- **Discounts Influence Purchasing Behavior**

Promotional strategies play an important role in driving purchases, especially for certain products.

- **Customer Engagement Varies Across Segments**

Customers with higher purchase history demonstrate stronger engagement, indicating opportunities for loyalty programs.

7. Business Recommendations

Based on the findings, the following recommendations are proposed:

- **Expand Subscription Programs**

Develop targeted campaigns and incentives to convert non-subscribers into subscribers, increasing long-term customer value.

- **Strengthen Customer Retention Strategies**

Implement loyalty programs and personalized offers for highly engaged customers to maintain and enhance retention.

- **Optimize Discount Strategy**

Use data-driven approaches to apply discounts more effectively, ensuring they drive sales without significantly reducing profit margins.

- **Focus on High-Performing Categories**

Allocate more resources to marketing and inventory for top-performing product categories to maximize revenue growth.

- **Implement Targeted Marketing**

Leverage demographic insights (such as age groups) to design more personalized and effective marketing campaigns.